MODULE 10A · EPISODE 1

Space Exploration: From Sputnik to Apollo and Beyond

Exploration and Flight History

STUDENT STUDY GUIDE

ASTRONOMY: OUR JOURNEY THROUGH THE COSMOS

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EPISODE OVERVIEW

Space Exploration: From Sputnik to Apollo and Beyond

This episode chronicles the dawn of the space age, from the first beep of Sputnik 1 in 1957 to the Voyager probes now exploring interstellar space. The panel traces the space race between the Soviet Union and the United States: the first satellites, the first humans in space, the tragedies and near misses along the way, and President Kennedy's challenge that culminated in Apollo 11's landing on the Moon. It then steps back to the aviation pioneers whose courage made spaceflight imaginable, and closes with the robotic probes that have served as our eyes and ears across the Solar System.

The Panel

Name	Role
Dr. Edward Thorne	Host; astrophysicist
Dr. Lena Petrova	Guest; exoplanet researcher
Dr. Kenji Tanaka	Guest; microbial ecologist

Learning Objectives

After this episode you should be able to:

- Explain why the launch of Sputnik 1 in 1957 began the space race, and describe the US response.
- Describe what Telstar 1 achieved and how radiation in the Van Allen Belts cut its life short.
- Distinguish orbital from suborbital flight, using Gagarin, Shepard, and Glenn as examples.
- Identify the first Soviet and American astronauts and the risks and scares of their missions.
- Explain the significance of the Apollo 1 fire and of Kennedy's 1962 Rice University speech.
- Describe the Apollo 11 landing, the roles of its three astronauts, and the problems they overcame.
- Identify the achievements of the Wright Brothers, Bessie Coleman, Charles Lindbergh, and Amelia Earhart.
- Summarize what the Pioneer, Viking, and Voyager probes revealed about the Solar System.
- Place the major events of early flight and spaceflight on a timeline.

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QUICK REFERENCE

Key Terms

Term	Definition
Space age	The era of human activity in space, which began with the launch of Sputnik 1 in 1957.
Space race	The competition between the Soviet Union and the United States to reach milestones in spaceflight first.
Artificial satellite	A human-made object placed in orbit around Earth or another body.
Sputnik 1	The first artificial satellite: a 23-inch metal sphere launched by the Soviet Union on October 4, 1957.
Vanguard TV-3	The first US attempt to launch a satellite; it failed at launch in December 1957.
Explorer 1	The first successful US satellite, launched January 31, 1958.
Telstar 1	The first transatlantic telecommunication satellite (1962); it relayed the first television images across the Atlantic.
Van Allen Belts	Zones of energetic charged particles trapped by Earth's magnetic field; first detected by Explorer 1.
Orbital flight	A flight fast enough to circle Earth continuously, falling around the planet rather than back to it.
Suborbital flight	A flight that reaches space but lacks the speed to circle Earth, so it arcs back down.
Reentry	The return of a spacecraft through Earth's atmosphere.
Heat shield	The protective layer on a capsule that absorbs and carries away the intense heat of reentry.
Vostok 1	Yuri Gagarin's capsule, which made the first human orbit of Earth in 1961.
Liberty Bell 7	Gus Grissom's capsule; it sank after its hatch blew prematurely.
Friendship 7	John Glenn's capsule, which made the first American orbits of Earth in 1962.
Apollo program	NASA's program to land astronauts on the Moon; six missions landed between 1969 and 1972.
Lunar module	The craft that carried two astronauts from lunar orbit to the surface and back; Apollo 11's was the Eagle.
Command module	The crew capsule that stayed in lunar orbit and returned the crew to Earth; Apollo 11's was Columbia.
Heavier-than-air flight	Flight by a craft denser than air, held up by wings and engine power rather than by buoyancy as a balloon is.
Barnstormer	A stunt pilot who toured the country giving air shows and rides, especially in the 1920s.
Spirit of St. Louis	Charles Lindbergh's single-engine airplane, flown solo from New York to Paris in 1927.
Unmanned (robotic) probe	An uncrewed spacecraft sent to explore other worlds and radio its findings back to Earth.

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Term	Definition
Interstellar space	The space between the stars, beyond the bubble of solar wind that surrounds the Sun.

QUICK REFERENCE

Numbers to Know

Value	What it refers to
October 4, 1957	Launch of Sputnik 1; the space age begins
23 inches (about 58 cm)	Diameter of the Sputnik 1 sphere
18,000 mph (about 29,000 km/h)	Sputnik 1's speed in orbit
January 31, 1958	Explorer 1, the first successful US satellite
1962	Telstar 1 launched, about four years after Explorer 1
4 months	Telstar 1's working life before radiation damage stopped it
Early 1963	Telstar 1 goes silent for good
April 12, 1961	Yuri Gagarin becomes the first human to orbit Earth
50%	Estimated chance of success for Gagarin's mission
May 5, 1961	Alan Shepard's suborbital flight, the first American in space
February 20, 1962	John Glenn becomes the first American to orbit Earth
3	Orbits completed by Glenn in Friendship 7
1962	Kennedy's "We choose to go to the Moon" speech at Rice University
1967	Apollo 1 fire kills Grissom, White, and Chaffee
77	Glenn's age when he returned to space aboard the Space Shuttle
July 20, 1969	Apollo 11's lunar module, the Eagle, lands on the Moon
Less than 1 minute	Fuel to spare when the Eagle touched down
12 men, 6 missions	Moonwalkers across the Apollo landings
1903	The Wright Brothers' first powered, controlled flight at Kitty Hawk
1927	Lindbergh's solo flight from New York to Paris
33 hours	Approximate length of Lindbergh's flight
1937	Amelia Earhart disappears attempting to fly around the globe
1976	Viking missions send the first clear images from the surface of Mars
1977	Launch of the twin Voyager spacecraft
4	Giant planets visited by Voyager 2: Jupiter, Saturn, Uranus, and Neptune
Nearly 50 years	How long both Voyagers have been operating

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EPISODE CONTENT

Part-by-Part Summary

Part 1 — Sputnik and the Start of the Space Race

The space age began on October 4, 1957, when the world awoke to the news that the Soviet Union had launched **Sputnik 1**, the first artificial satellite. It was a 23-inch (about 58 cm) metal sphere orbiting Earth at about **18,000 miles per hour**, and its simple radio beep signaled that humanity had crossed a new frontier. The **space race** was officially on.

The US response was hurried. Its first attempt, **Vanguard TV-3**, failed. Then, on **January 31, 1958**, **Explorer 1** successfully reached orbit, becoming the first US satellite.

Part 2 — Telstar 1 and the Van Allen Belts

About four years later came the real leap for the US: **Telstar 1**, the first transatlantic telecommunication satellite. Launched in 1962, it relayed the **first television images across the Atlantic**, including a press conference by President Kennedy.

Its life was cut short. Telstar 1 suffered radiation damage while passing through the **Van Allen Belts**, zones of charged particles held by Earth's magnetic field, which had been intensified by a high-altitude US nuclear test. It failed just **four months** after launch and, after a brief revival, went silent for good in early 1963.

Part 3 — The First Humans in Space

The race then shifted to putting a human in orbit, and the Soviets once again took the lead. On **April 12, 1961**, **Yuri Gagarin** became the first human to orbit the Earth. The mission was incredibly risky, with only about a **50% chance of success**. His **Vostok 1** capsule tumbled violently on reentry, but he ejected as planned and parachuted safely to the ground.

The US followed less than a month later with **Alan Shepard** on **May 5, 1961**. His was a **suborbital** flight: it reached space but did not have the speed to circle Earth, so it arced back down. It was still a crucial step, and Shepard would later walk on the Moon as commander of **Apollo 14**.

Gus Grissom was the second American in space. His flight was successful, but his capsule, the **Liberty Bell 7**, sank after its hatch blew prematurely. Grissom's story has a tragic end: in 1967 he, **Ed White**, and **Roger Chaffee** died in a fire during a launch pad test for **Apollo 1**. The devastating loss led to major safety improvements in the Apollo program.

> *† The episode says the hatch "blew prematurely." It did, but the cause was never determined. Grissom always said he did not trigger it, and investigators did not fault him.*

Part 4 — Kennedy's Challenge and John Glenn

The driving force behind the American effort was **President Kennedy**. His **1962 speech at Rice University** is legendary: "We choose to go to the Moon in this decade and do the other things, not because they are easy, but because they are hard." That speech galvanized the nation and NASA.

Earlier that same year, **John Glenn** had already given the nation a boost. On **February 20, 1962**, he became the **first American to orbit the Earth**, completing **three orbits** in his

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Friendship 7 capsule. Like Gagarin, he had a scare: a faulty signal suggested his **heat shield** was loose. He made it back safely, went on to a distinguished career as a US Senator, and even returned to space at **age 77** aboard the Space Shuttle.

† Kennedy first set the Moon goal in May 1961, in an address to Congress. The 1962 Rice speech explained the goal to the public and rallied support for it.

Part 5 — Apollo 11 and the Moon Landings

All of this culminated in the summer of 1969. On **July 20, 1969**, the world watched as Apollo 11's **lunar module**, the **Eagle**, touched down on the Moon. **Neil Armstrong** and **Buzz Aldrin** became the first humans to walk on another world, while **Michael Collins** orbited above in the **command module Columbia**. It fulfilled Kennedy's challenge, a monumental achievement for humanity.

The landing was fraught with tension. Armstrong had to take **manual control** to avoid a boulder field, landing with **less than a minute of fuel** to spare. On departure, a critical circuit breaker broke, and the crew had to improvise with a pen to get the engine to fire.

In the end, **12 men walked on the Moon across six Apollo missions**. It remains one of the greatest accomplishments in human history.

Part 6 — Pioneers of Flight

Spaceflight stands on the shoulders of early aviation pioneers. The **Wright Brothers** achieved the first **powered, heavier-than-air controlled flight** at Kitty Hawk in **1903**.

Bessie Coleman was the first African-American and Native-American woman to earn a pilot's license. Denied the opportunity in the US, she went to France to learn to fly and became a successful **barnstormer**.

In **1927**, **Charles Lindbergh** flew the **Spirit of St. Louis** solo from New York to Paris, a grueling **33-hour** flight that made him a global hero. **Amelia Earhart** became the first woman to fly solo across the Atlantic. Her attempt to circumnavigate the globe in **1937** ended in her mysterious disappearance, a tragedy that still captures the public imagination.

† Lindbergh's flight was the first solo nonstop crossing of the Atlantic. Others had already flown it nonstop with a two-person crew (Alcock and Brown, 1919).

Part 7 — Robotic Explorers

Alongside the crewed missions, **unmanned probes** have been our robotic eyes and ears in the cosmos. **Pioneer 10 and 11** gave us our first close-up images of **Jupiter and Saturn**.

The **Viking** missions in **1976** provided the first clear images from the surface of **Mars** and conducted experiments that some scientists thought hinted at living microbes, though most now credit ordinary soil chemistry.

The twin **Voyager** spacecraft, launched in **1977**, made a grand tour of the outer planets. **Voyager 2** visited **Jupiter, Saturn, Uranus, and Neptune**, and both Voyagers are still operating nearly 50 years later, exploring **interstellar space**.

From the first beep of Sputnik to the Voyager probes pushing the boundaries of our Solar System, the story of space exploration is one of courage, ingenuity, and an unyielding desire to reach for the stars.

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TRY IT YOURSELF

Race Lindbergh, then lap the planet. Lindbergh flew about 3,600 miles (about 5,800 km) from New York to Paris in about 33 hours. What was his average speed in miles per hour? Sputnik 1 orbited at about 18,000 mph. About how many times faster was it? At that speed, about how long would it take to travel once around Earth (circumference about 25,000 miles, or 40,000 km)?

(Answer: 3,600 miles ÷ 33 hours ≈ 110 mph, about 175 km/h. Sputnik was 18,000 ÷ 110 ≈ 160 times faster. Once around Earth: 25,000 miles ÷ 18,000 mph ≈ 1.4 hours, about 83 minutes. Sputnik actually took about 96 minutes per orbit, because its orbit was a larger circle, hundreds of miles above the ground.)

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STUDY QUESTIONS

Questions for Review

Answer in your own words. Where a question asks you to explain rather than to state, a one-sentence answer is unlikely to be complete.

Red numbers mark core questions — the ones to be sure you can answer.

Part 1 — Sputnik and the Start of the Space Race

1. What happened on October 4, 1957, and why is that date considered the start of the space age?

2. Describe Sputnik 1: its shape, its size, and how fast it traveled.

3. How did the United States respond to Sputnik? Name its failed first attempt and its first successful satellite, with the date.

4. What was the space race?

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Part 2 — Telstar 1 and the Van Allen Belts

1. What was Telstar 1, and what historic first did it achieve?

2. What damaged Telstar 1, and what had made that hazard worse?

3. How long did Telstar 1 work before failing, and when did it go silent for good?

Part 3 — The First Humans in Space

1. Who was the first human to orbit the Earth? Give the date and the name of the capsule.

2. Why was Gagarin’s mission considered so risky, and what happened during his return to Earth?

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3. How was Alan Shepard's flight different from Gagarin's? Explain the difference between orbital and suborbital flight.

4. What did Shepard go on to do later in his career?

5. Who was the second American in space, and what happened to his capsule?

6. What happened during the Apollo 1 launch pad test in 1967, and what was its lasting effect?

Part 4 — Kennedy's Challenge and John Glenn

1. What did President Kennedy say in his 1962 speech at Rice University, and what effect did the speech have?

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2. Who was the first American to orbit the Earth? Give the date, the name of the capsule, and the number of orbits.

3. What scare did Glenn face during his mission?

4. What did Glenn do after his orbital flight?

Part 5 — Apollo 11 and the Moon Landings

1. When did Apollo 11 land on the Moon? Name the lunar module and the command module.

2. Name the three Apollo 11 astronauts and describe each one’s role.

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3. Describe two problems the Apollo 11 crew had to overcome.

4. How many people have walked on the Moon, and across how many missions?

5. In what way did Apollo 11 fulfill Kennedy's challenge?

Part 6 — Pioneers of Flight

1. What did the Wright Brothers achieve, and where and when did they do it?

2. Who was Bessie Coleman, and why did she go to France?

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3. What did Charles Lindbergh accomplish in 1927?

4. What "first" did Amelia Earhart achieve, and what happened in 1937?

5. Why does the episode say that spaceflight "stands on the shoulders" of early aviation pioneers?

Part 7 — Robotic Explorers

1. What is an unmanned probe, and why does the episode call probes our "robotic eyes and ears"?

2. What did the Pioneer 10 and 11 missions give us?

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3. What did the Viking missions accomplish in 1976? What did some scientists think their experiments showed, and what do most scientists think now?

4. When were the Voyager spacecraft launched, which planets did Voyager 2 visit, and where are the Voyagers now?

5. According to the episode, what three qualities define the story of space exploration?

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YOUR NOTES

Notes

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